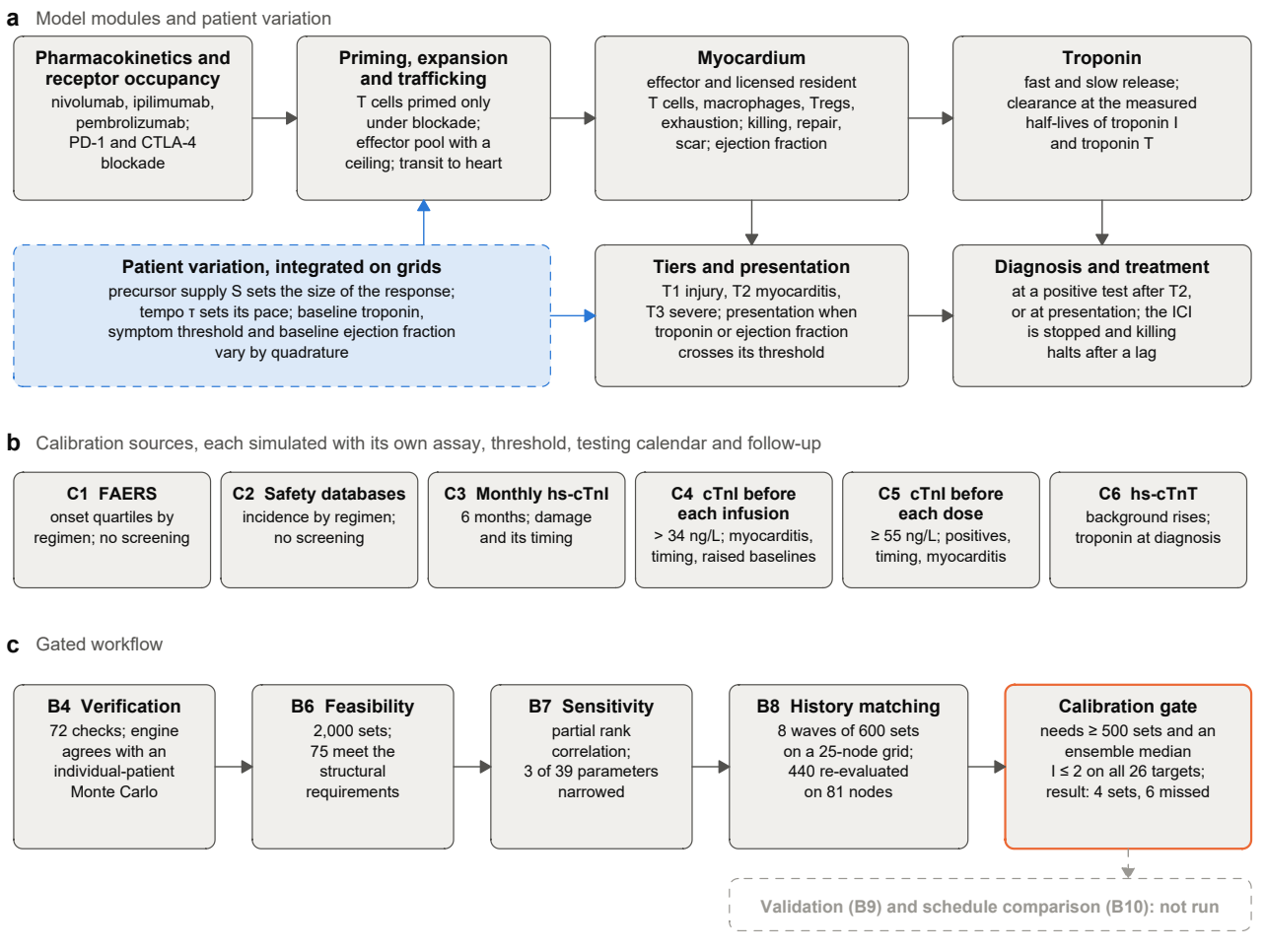

a
Model modules and patient variation
Pharmacokinetics and receptor occupancy
nivolumab, ipilimumab, pembrolizumab; PD-1 and CTLA-4 blockade
Priming, expansion and trafficking
T cells primed only under blockade; effector pool with a ceiling; transit to heart
Myocardium
effector and licensed resident T cells, macrophages, Tregs, exhaustion; killing, repair, scar; ejection fraction
Troponin
fast and slow release; clearance at the measured half-lives of troponin I and troponin T
Patient variation, integrated on grids
precursor supply S sets the size of the response; tempo τ sets its pace; baseline troponin, symptom threshold and baseline ejection fraction vary by quadrature
Tiers and presentation
T1 injury, T2 myocarditis, T3 severe; presentation when troponin or ejection fraction crosses its threshold
Diagnosis and treatment
at a positive test after T2, or at presentation; the ICI is stopped and killing halts after a lag
b
Calibration sources, each simulated with its own assay, threshold, testing calendar and follow-up
C1 FAERS
onset quartiles by regimen; no screening
C2 Safety databases
incidence by regimen; no screening
C3 Monthly hs-cTnI
6 months; damage and its timing
C4 cTnI before each infusion
> 34 ng/L; myocarditis, timing, raised baselines
C5 cTnI before each dose
≥ 55 ng/L; positives, timing, myocarditis
C6 hs-cTnT
background rises; troponin at diagnosis
c
Gated workflow
B4 Verification
72 checks; engine agrees with an individual-patient Monte Carlo
B6 Feasibility
2,000 sets; 75 meet the structural requirements
B7 Sensitivity
partial rank correlation; 3 of 39 parameters narrowed
B8 History matching
8 waves of 600 sets on a 25-node grid; 440 re-evaluated on 81 nodes
Calibration gate
needs ≥ 500 sets and an ensemble median I ≤ 2 on all 26 targets; result: 4 sets, 6 missed
Validation (B9) and schedule comparison (B10): not run